

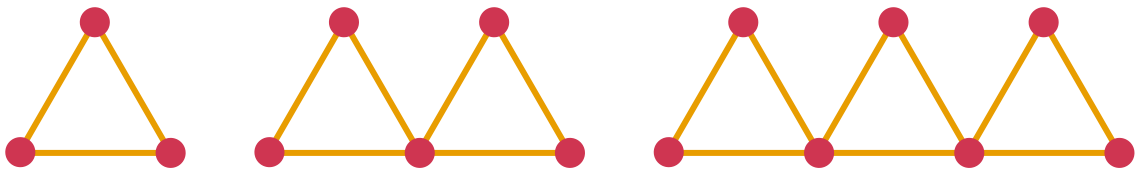
Tables and graphs



1 A diver starts at the surface of the water and starts to descend below the surface at a constant rate. The table below shows the depth of the diver over 4 minutes:

Number of minutes passed (x)	0	1	2	3	4
Depth of diver in metres (y)	0	1.35	2.7	4.05	5.4

- a Write an equation that describes the relationship between the number of minutes passed (x) and the depth (y) of the diver.
 - b At what depth would the diver be after 59 minutes?
 - c What does the numerical coefficient in the equation from part (a) represent?
- 2 Matches were used to make the pattern below:



- a Complete the table below:
- | | | | | | | |
|-----------------------------|---|---|---|---|----|----|
| Number of triangles (t) | 1 | 2 | 3 | 5 | 10 | 20 |
| Number of matches (m) | | | | | | |
- b Write an equation that describes the relationship between the number of matches (m) and the number of triangles (t).
 - c Hence, how many matches are required to make 21 triangles?

- 3 A racing car starts the race with 80 litres of fuel.  From there, it uses fuel at a rate of 2 litres per minute.

a Complete the table below:

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in tank (y)					

- b Write an algebraic relationship linking the number of minutes passed (x) and the amount of fuel left in the tank (y).
- c After how many minutes, x , will the car need to refuel (i.e. when there is no fuel left) ?
- d Explain why plotting the points from the table in part (a) results in a straight line graph.
- 4 There are 40 litres of water in a rainwater tank. It rains for a period of 24 hours and during this time, the tank fills up at a rate of 10 litres per hour.

a Complete the table below:

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

- b Write an algebraic relationship linking the number of hours passed (x) and the amount of water in the tank (y).
- c Sketch the graph of the linear equation from part (b).
- 5 It starts raining and an empty rainwater tank fills up at a constant rate of 2 litres per hour. By midnight, there are 10 litres of water in a rainwater tank. As it rains, the tank continues to fill up at this rate.

a Complete the table below:

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

- b Write an algebraic relationship linking the number of hours passed since midnight (x) and the amount of water in the tank (y).
- c How many hours before midnight was the tank empty (i.e. when $y = 0$)? Note that x represents the number of hours passed since midnight, so $-x$ represents x hours before midnight.
- d Sketch the graph of the linear equation from part (b).

6 Consider the table below:



Time in minutes (x)	1	2	3	4	5
Temperature in $^{\circ}\text{C}$ (y)	5	9	13	17	21

- a By how much is the temperature increasing each minute?
- b What would the temperature have been at time 0 minutes?
- c Find the algebraic relationship between x and y .
- d Sketch the graph of the linear equation from part (c).

7 In a study, scientists found that the more someone sleeps, the quicker their reaction time. The attached table shows the findings:

Number of hours of sleep (x)	0	1	2	3	4	5
Reaction time in seconds (y)	9	8.8	8.6	8.4	8.2	8

- a How much does the reaction time change for each extra hour of sleep?
- b Write an algebraic equation relating the number of hours of sleep (x) and the reaction time (y).
- c Find the reaction time for someone who has slept 1.5 hours.
- d Find the number of hours someone sleeps if they have a reaction time of 8.5 seconds.

Word problems

8 Consider the statement: 9 divided by the sum of x and 3 is equal to 6 divided by the difference between x and 3.

- i Write an equation for the statement.
- ii Solve the equation for x .

9 Eileen tries to guess how many people are at a concert, but she guesses 400 too many. Adam guesses 250 too few. The average of their guesses is 2575. Set up an equation and solve for x , the exact number of people at the concert.

10 Kate and Isabelle do some fundraising for their sporting team. Together they raised \$600. If Kate raised \$272 more than Isabelle, and Isabelle raised \$ p :

- a Write an equation in terms of p that represents the relationship between the different amounts and solve for p .



- 11** Caitlin's phone plan allows her to send 300 text messages for free. But if she sends over 300 messages, the cost, C , in dollars of sending x text messages is modelled by:

$$C = 0.1(x - 300) + 8$$

Solve for the number of text messages x that have been sent when the total cost is \$30.10.

- 12** To manufacture doors, the manufacturer has a fixed cost of \$4200 plus a variable cost of \$100 per door. Find n , the number of doors that need to be produced so that the average cost per door is \$120.
- 13** Eileen's sink can be filled in 8 minutes and can subsequently be drained in 16 minutes when it is unplugged. Find the number of minutes, n , that it will take to fill the sink if it is left unplugged while being filled.
- 14** Oprah can paint the average house in 15 hours, while Tobias can do it in 10 hours. Find the number of hours, t , that it would take to paint a house if they both worked together.
- 15** A phone salesperson earned \$880 in a particular week during which she sold 22 units. In another week she made \$925 and sold 25 units. Let x be the number of units sold and y be the weekly earnings.
- a** Find the linear equation that models the units-earnings relationship for this salesperson.
 - b** Hence, predict the earnings of the salesperson if she sells 36 units.
- 16** Sisters, Ursula and Eileen, are training for a triathlon event. Ursula finds that her average cycling speed is 13 km/h faster than Eileen's average running speed. Ursula can cycle 46 km in the same time that it takes Eileen to run 23 km.
- a** If Eileen's running speed is n km per hour, solve for n .
 - b** Find Ursula's average cycling speed.
- 17** When a new car was released on the market, car journalists, Susana and Irene, were invited to take the car for a test drive. Each took their car into the mountains, where Susana covered a distance of 31 km in the same time that Irene drove 25 km. Susana drove 12 km/h faster than Irene.
- a** If Irene drove at a speed of n km/h, solve for n .
 - b** Hence, find Susana's speed.

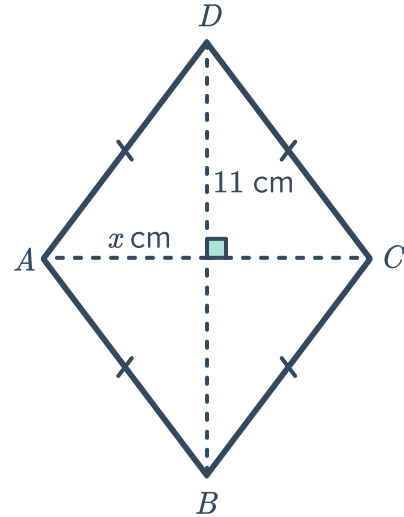
Geometric applications

18 Find the perpendicular height, h , of a parallelogram that has an area of 66 cm^2 and a base of 6 cm. The area of a parallelogram is given by: $A = bh$.

19 Find the base length, b , of a parallelogram that has an area of 228 mm^2 and a perpendicular height of 12 mm. The area of a parallelogram is given by: $A = bh$.

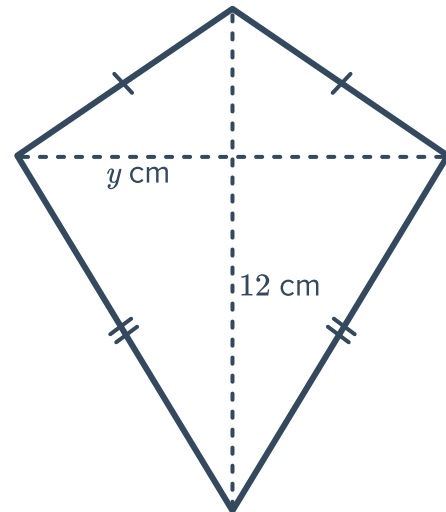
20 Find the missing length, x , if the area of rhombus $ABCD$ has an area of 44 cm^2 . The area of a rhombus is given by:

$$A = \frac{1}{2}xy$$



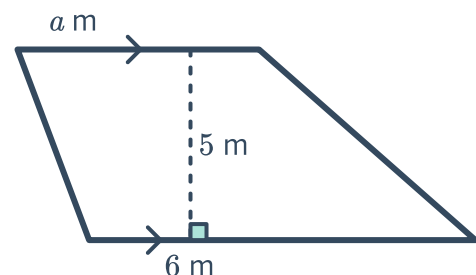
21 Find the missing length, y , if the area of the kite is 48 cm^2 . The area of a kite is given by:

$$A = \frac{1}{2}xy$$

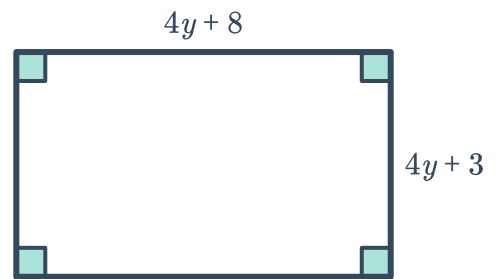


22 Find the missing side length, a , if the area of the trapezium shown is 25 m^2 . The area of a trapezium is given by:

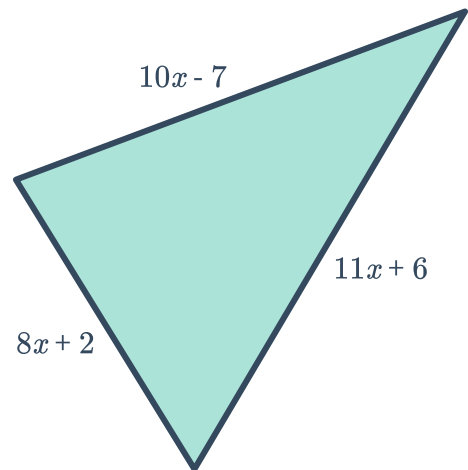
$$A = \frac{1}{2}(a + b)h$$



- 23** The perimeter of the rectangle is $126 + 3y$ cm. Solve for y .



- 24** The perimeter of the triangle is 146 cm. Solve for x .



- 25** The perimeter of the quadrilateral is 315 cm. Solve for x .

